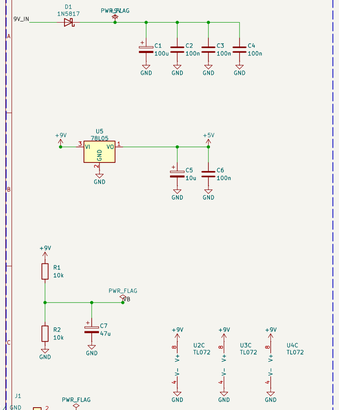
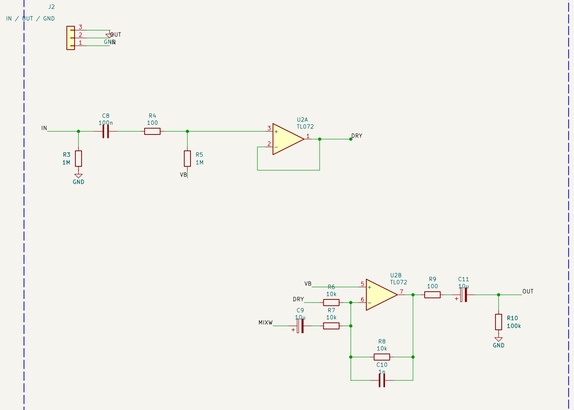
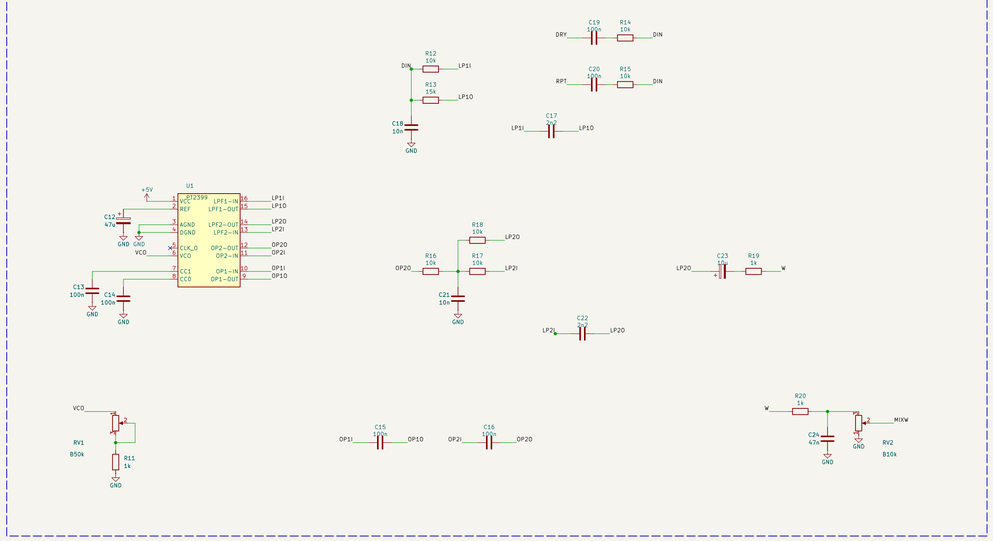
POWER: 9 V, VB 4.5 V for the TL072s, 5 V regulator for the PT2399



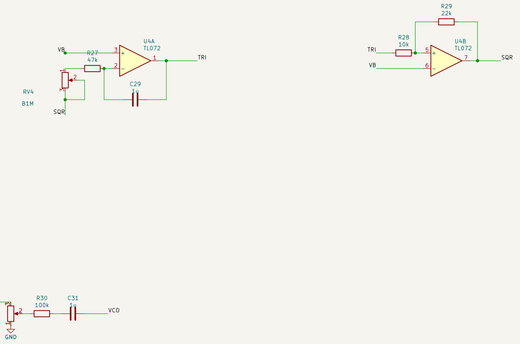
INPUT BUFFER (U2A) + OUTPUT MIXER (U2B): as Small Time



DELAY: PT2399 (U1), network as Small Time. TIME = R11 1k + B50k (VCO to ground)

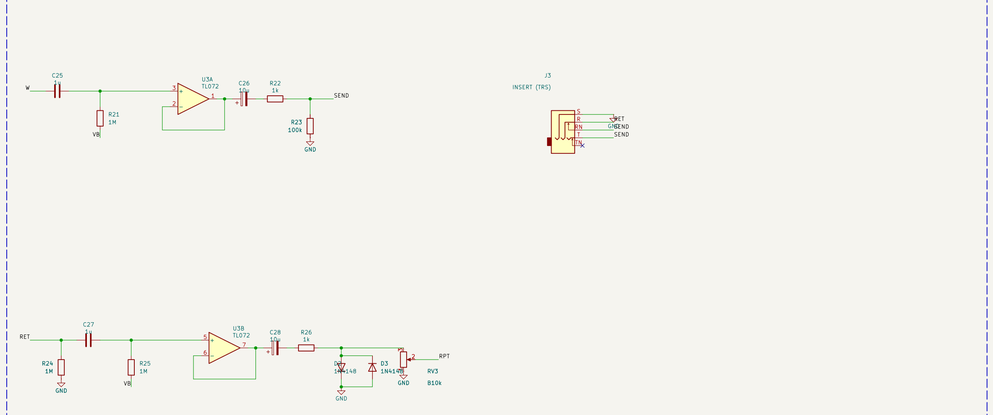


MODULATION: LFO (U4) -> DEPTH -> VCO pin (wow / chorus on the echoes). ~0.5–12 Hz



The LFO current into pin 6 swings the clock: about +–15 uA at full DEPTH. Compare with the mod examples.

FEEDBACK LOOP: SEND (U3A) -> TRS insert -> RETURN (U3B) -> limiter -> REPEAT. No plug: ring–normal ties RETURN to SEND



Only the repeats circulate: the first echo is clean, each later echo goes through the loop again.

A fuzz in the loop adds gain every pass: the 1N4148s cap the loop at about +–0.6 V so oscillation stays musical.

PT2399 network from Small Time: true bypass
Single PT2399, effects loop in the feedback path (TRS insert), TIME modulation

Super-Freq

Sheet: /
File: LoopEcho.kicad_sch

Title: Loop Echo (breadboard)

Size: A1 Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.4	Id: 1/1